Chapter 1: Data Mining Project Methodology

<{http://www.bookeducator.com/Textbook}learningobjective >Students will be able to explain the six iterative phases of CRISP-DM and describe how data influences every phase of the process <{http://www.bookeducator.com/Textbook}learningobjective >Students will be able to evaluate project feasibility by assessing practical impact, data availability, and analytical feasibility <{http://www.bookeducator.com/Textbook}learningobjective >Students will be able to distinguish between decision support (explanatory) and machine learning pipeline (predictive) project types and select appropriate approaches <{http://www.bookeducator.com/Textbook}learningobjective >Students will be able to compare CRISP-DM with alternative project methodologies (TDSP, SEMMA, KDD, OSEMN) and identify strengths and limitations of each

1.1Introduction

Making Sense of the Pieces: CRISP-DM

One of the most common questions students ask is, “How does everything we’re learning fit together, and why are we doing it?” A practical way to address this question is by introducing a framework that is widely used in real-world data projects: the **Cross-Industry Standard Process for Data Mining (CRISP-DM)**.

This download can be found online.

CRISP-DM A methodology for understanding how business problems are solved using aria-live="polite" aria-atomic="true"> provides a structured, iterative approach for translating business objectives into data-driven outcomes. Although the tools and techniques used in data analytics continue to evolve, this framework remains relevant because it emphasizes adaptable principles rather than rigid, tool-specific procedures.

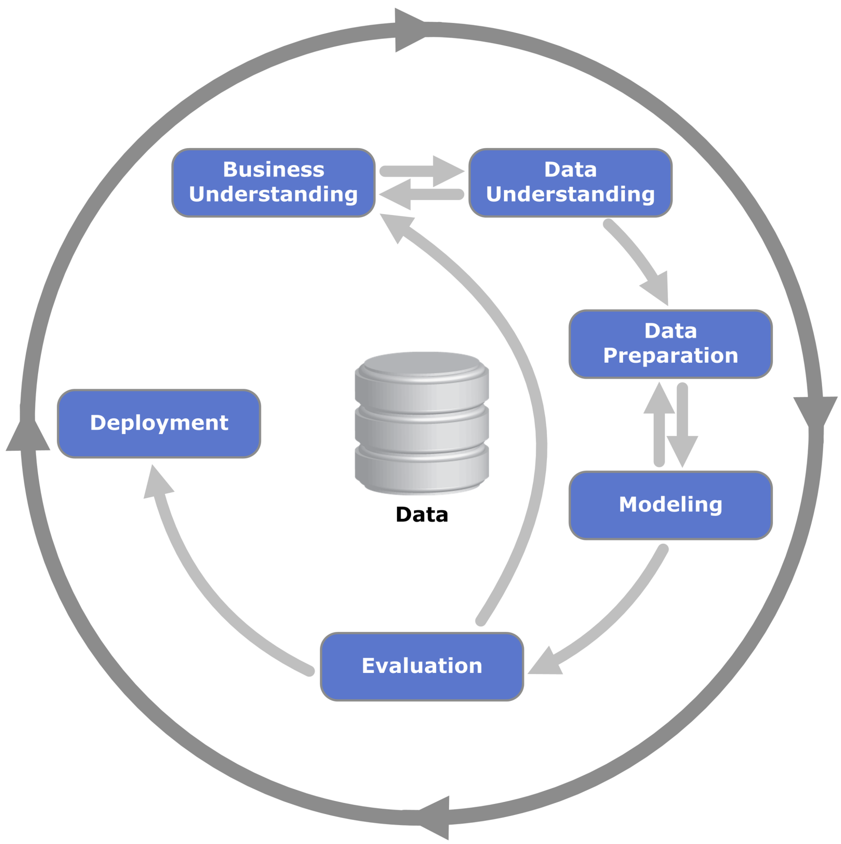


Figure 1.1: CRISP-DM Process

The CRISP-DM Cycle

At its core, CRISP-DM is an iterative cycle with data at the center, emphasizing the role data plays in every phase of a project. The framework consists of six interconnected phases:

1. **Business Understanding:** Define the business problem, objectives, and success criteria.
2. **Data Understanding:** Identify available data sources, explore the data, and assess data quality.
3. **Data Preparation:** Clean, transform, and organize data so it is suitable for analysis or modeling.
4. **Modeling:** Select modeling techniques, algorithms, and features appropriate for the problem.
5. **Evaluation:** Evaluate model performance relative to business objectives and baseline approaches.
6. **Deployment:** Deliver results to stakeholders or integrate the solution into operational systems.

Importantly, CRISP-DM is not a linear checklist. Teams frequently revisit earlier phases as new insights emerge, data limitations are discovered, or business objectives evolve.

Characteristics of CRISP-DM

1. **Data Is Central:** Data influences every phase of the process, from shaping business questions to constraining modeling choices and deployment options.

2. **Iterative, Not Linear:** Data projects rarely proceed in a straight line. You often do not know whether the available data can support your objectives until you begin exploring and experimenting.
3. **"Satisficing A decision-making concept in which an option is selected not because it is optimal, but because the additional cost of improving it outweighs the additional value gained." Over Perfection:** While accuracy is important, there is a point at which additional refinements no longer justify their time and cost. Data scientists must balance technical improvements with business value.

In practice, this means that although data scientists may revisit nearly every phase multiple times, they are not expected to achieve perfection. At some point, a model must be selected and deployed. A model that is slightly less accurate but timely and actionable may be far more valuable than a marginally better model that arrives too late or costs too much to develop.

Flattening the Cycle: Questions for Each Phase

Another way to understand CRISP-DM is by focusing on the key questions each phase is designed to answer. This question-based view helps clarify the purpose of each phase, especially for those new to data projects.

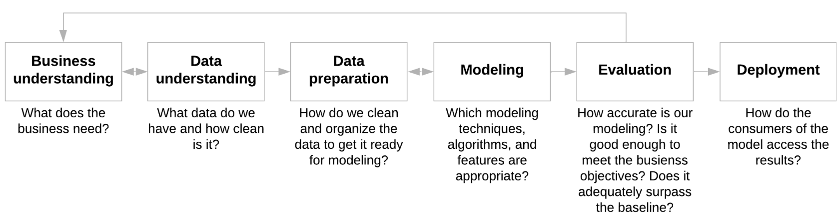


Figure 1.2: CRISP-DM Diagram

Together, the process-based and question-based views reinforce that CRISP-DM is a flexible framework intended to guide thinking, not prescribe rigid steps.

Putting It Into Practice

Throughout this course, you have already been applying the CRISP-DM process, even if it was not explicitly labeled. Exploratory data analysis projects, dashboards, and predictive modeling assignments all naturally map to its phases.

The goal is for this framework to become second nature, ensuring that you always understand what you are doing, why it matters, and how each step contributes to a successful outcome. In the following sections, we examine each phase in greater detail and later compare CRISP-DM to alternative project methodologies.

1.2Phase 1: Business Understanding

Understanding the Business Problem

The Business Understanding phase is critical for identifying opportunities where data mining can address business problems. Organizations often begin with a clear sense of potential opportunities, but many of the most successful companies intentionally dedicate time to brainstorming creative and high-impact possibilities before committing technical resources.

What Makes a Successful Data Project?

This phase defines the scope of the project by evaluating several key criteria that determine whether a data project is likely to succeed:

1. **Business Problem:** What specific challenge or opportunity is being addressed, and why does it matter?
2. **Data Availability:** Do we have the necessary data, or can it be feasibly acquired at a reasonable cost?
3. **Analytical Feasibility:** Can the available data support reliable analysis or prediction using appropriate statistical or machine learning methods?
4. **Process Integration:** Can the results be reasonably integrated into existing systems, workflows, or decision-making processes?
5. **Operational Disruption:** Will the project introduce inefficiencies or disruptions, particularly in sensitive environments such as healthcare or emergency services?

While all of these criteria are important, the first three primarily fall within the analytical responsibilities of the data scientist. The final two are organizational constraints that must be considered before a project moves forward. The diagram below focuses on the analytical core of project feasibility.

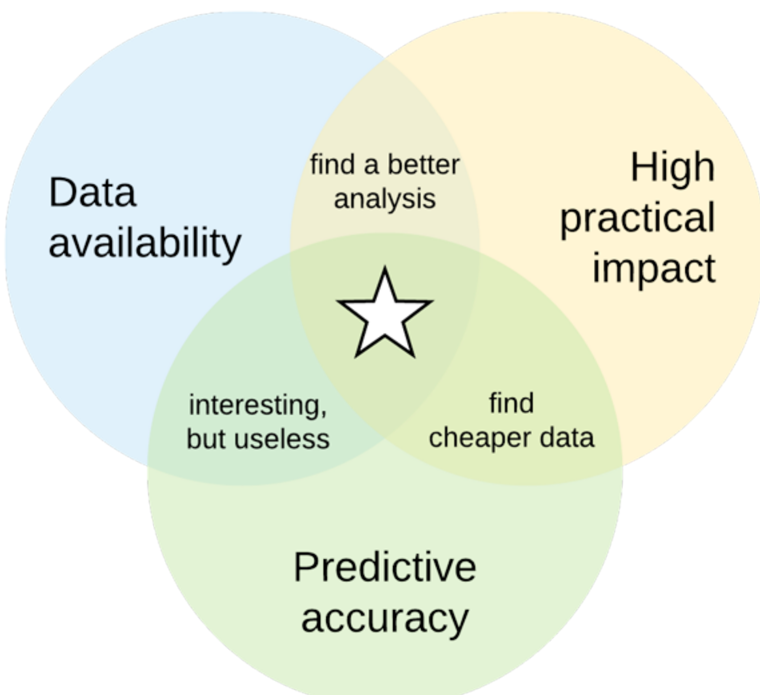


Figure 1.3: Identifying Opportunities for Data Mining

The yellow circle, labeled "**high practical impact**," represents projects that address meaningful business problems. Data projects typically begin here, as even technically sophisticated solutions provide little value if they do not address an important organizational need.

The blue circle, labeled "**data availability**," highlights the importance of accessible and affordable data. When relevant data already exists within organizational systems, costs are typically low. However, projects that rely on third-party data purchases, extensive surveys, or large-scale web scraping may become infeasible if acquisition costs exceed expected benefits.

The green circle, labeled "**analytical feasibility**," reflects whether the available data can support reliable analysis or prediction. For predictive modeling projects, this often means achieving sufficient predictive accuracy. For non-predictive projects, such as exploratory analysis or dashboards, feasibility may instead depend on interpretability, stability, or decision support value.

A data scientist leading a data project must ensure that these three analytical criteria are met to a reasonable degree before advancing to later phases.

Example: Nurse Scheduling in a Healthcare Setting

A healthcare provider sought to improve nurse scheduling by predicting required nurse hours in an emergency room. Data collection involved tracking two indicators: whether lab results were abnormal and the charge nurse's subjective assessment of case complexity. Initial analysis suggested a potential 60 percent improvement in predictive performance.

- **Data Collection Cost:** Nurses were required to manually enter data into a separate system.
- **Workflow Disruption:** Even a streamlined mobile application proved too disruptive in a fast-paced emergency room environment.

Despite promising analytical results, the project was discontinued because the operational costs and workflow disruptions outweighed the expected benefits. This example illustrates how a project can fail during the Business Understanding phase even when predictive performance appears strong.

Deliverables of the Business Understanding Phase

The Business Understanding phase resembles the initiation stage of traditional project management and requires a careful feasibility analysis. Typical deliverables include:

1. **Project Scope and Objectives:** Clearly define the problem, goals, and success criteria, such as increasing sales, reducing costs, or improving retention.
2. **Feasibility Analysis:** Evaluate whether the project can be realistically executed given organizational constraints.
    - **Resource Inventory:** Assess the availability of data, expertise, computing resources, and software.
    - **Requirements and Constraints:** Identify scheduling, legal, ethical, security, or compliance considerations.
    - **Risk Assessment:** Identify potential risks and develop contingency plans.

3. **Cost-Benefit Analysis:** Compare anticipated benefits to both direct project costs and the opportunity cost of alternative initiatives.
4. **Project Plan:** Develop a high-level plan that outlines tasks, dependencies, timelines, resources, and expected iteration using appropriate project management practices.

Why the Business Understanding Phase Matters

The Business Understanding phase is foundational to CRISP-DM because it ensures that data projects are valuable, feasible, and aligned with organizational goals. By rigorously defining objectives and constraints at the outset, organizations reduce the risk of investing in technically sound projects that ultimately fail to deliver business value.

Related Articles

- 1.3Phase 2: Data Understanding

Exploring the Data: The Data Understanding Phase

The data understanding phase The second phase of the CRISP-DM, which focuses on initial data collection and activities designed to help you become familiar with the data, identify data quality problems, discover early insights, and detect interesting subsets for hypothesis generation., often referred to as exploratory data analysis (EDA) The process of performing initial data investigations to discover patterns, spot anomalies, test hypotheses, and check assumptions using summary statistics and visualizations., centers on learning what the data contains and what it can support. This phase builds directly on the Business Understanding phase and provides the analytical foundation for later data preparation and modeling.

Key Objectives of the Data Understanding Phase

1. **Understand the Data:** Become familiar with the dataset's structure, variables, scale, and limitations.
2. **Identify Target Variables:** Identify the target variable (also called the dependent variable or label) that represents the outcome of interest, such as sales, employee turnover, or patient costs.
3. **Analyze Relationships:** Explore how input variables (features) relate to the target variable using summary statistics and visualizations.
4. **Detect Quality Issues:** Identify missing, inconsistent, or anomalous data values and document potential remediation strategies.

It is important to note that the Data Understanding phase focuses on identifying data quality issues, not fixing them. Most data cleaning and transformation activities occur in the Data Preparation phase.

Example: Health Insurance Dataset Exploration

Consider the snapshot of a health insurance dataset shown below, which contains demographic and behavioral variables used to analyze medical insurance charges.

	A	B	C	D	E	F	G
1	age	sex	bmi	children	smoker	region	charges
2	19	female	27.9	0	yes	southwest	16884.924
3	18	male	33.77	1	no	southeast	1725.5523
4	28	male	33	3	no	southeast	4449.462
5	33	male	22.705	0	no	northwest	21984.47061
6	32	male	28.88	0	no	northwest	3866.8552

Figure 1.4: Snapshot of Health Insurance Dataset

A common first step in Data Understanding is generating univariate statistics to examine each variable independently. This helps analysts understand distributions, ranges, data types, and potential anomalies before examining relationships between variables.

Table 1.1
Sample Table of Descriptive Univariate Statistics

	Count	**Missing**	**Type**	**Min**	**Max**	**25%**	**50%**	**75%**	**Mean**	**Media**
Age	1338	0	int64	18	64	27	39	51	39.21	39
Sex	1338	0	object	—	—	—	—	—	—	—
BMI	1338	0	float64	15.96	53.13	26.3	30.4	34.69	30.66	30.4
Children	1338	0	int64	0	5	0	1	2	1.09	1
Smoker	1338	0	object	—	—	—	—	—	—	—
Region	1338	0	object	—	—	—	—	—	—	—
Charges	1338	0	float64	1121.87	63770.4	4740.29	9382.03	16639.9	13270.4	9382.03

From this table, analysts can quickly identify important characteristics of the data. For example, insurance charges are highly right-skewed, indicating the presence of extreme values that may influence modeling. The mix of numeric and categorical variables also signals the need for appropriate encoding and transformation strategies in later phases.

Further exploration may involve multivariate visualizations, such as a 3D scatterplot examining how insurance charges relate jointly to body mass index (BMI) and age. These visualizations help identify nonlinear patterns, interactions, and potential feature relationships.

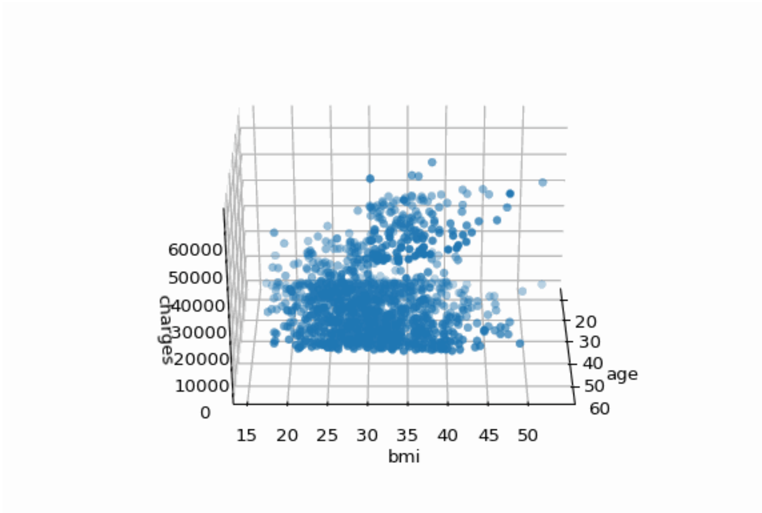


Figure 1.5: 3D Scatterplot Created During Data Understanding Phase

Importance of Data Understanding

Data Understanding plays a critical tactical role in a data project by revealing limitations, assumptions, and risks early. Without a solid grasp of the data, subsequent preparation and modeling decisions may be misguided or invalid.

Outputs of the Data Understanding Phase

1. **Initial data collection report**: Document data sources, acquisition methods, encountered issues, and resolutions.
2. **Data description report**: Summarize dataset structure, size, variable types, and surface-level characteristics.
3. **Data exploration report**: Present early insights, hypotheses, and supporting visualizations.
4. **Data quality report**: Identify missing or inconsistent data and document recommended remediation strategies.

Together, these outputs directly inform the Data Preparation and Modeling phases by guiding cleaning decisions, feature selection, and modeling strategy.

This download can be found online.

Why the Data Understanding Phase (EDA) Matters

The Data Understanding phase establishes the analytical foundation for the entire project. By thoroughly exploring and documenting the dataset, analysts reduce downstream risk and improve the reliability of modeling results. As a general rule, time invested in understanding the data pays dividends throughout the remainder of the project.

Class Discussion

Suppose a dataset shows strong correlations between features and a target variable during the Data Understanding phase, but the data was collected using a process that may influence outcomes. Should the project proceed to modeling, or should concerns about data quality and potential bias stop the project early? Why?

1.4Phase 3: Data Preparation

Preparing the Data: The Data Preparation Phase

The data preparation phase The third phase of the CRISP-DM, which covers all activities used to construct the final dataset that will be fed into modeling tools. involves transforming raw data into a refined dataset suitable for analysis and modeling. This phase is inherently iterative and does not follow a strict sequence, as preparation decisions often depend on insights uncovered during Data Understanding. The goal is to produce a smaller, relevant, and computationally efficient dataset that supports accurate and reliable modeling.

Key Objectives of Data Preparation

1. **Selection:** Choose the most relevant tables, records, and attributes (features) while discarding unnecessary or redundant data.
2. **Transformation:** Convert data into formats better suited for analysis, such as encoding categorical variables or applying mathematical transformations.
3. **Cleaning:** Address data quality issues such as missing values, outliers, and skewed distributions.
4. **Efficiency:** Improve processing speed and reduce storage costs by optimizing data volume and structure.

Unlike the Data Understanding phase, which focuses on identifying data issues, the Data Preparation phase performs the actual modifications needed to make the data usable for modeling.

Example: Data Preparation in Action

Common data preparation activities include the following:

- Converting categorical variables (for example, “High,” “Medium,” and “Low”) into ordinal or indicator values.
- Applying non-linear transformations, such as logarithmic or exponential functions, to reduce skewness.
- Creating new features, such as a debt-to-income ratio, that capture more meaningful relationships than raw variables alone.

These steps are often performed manually during exploration and experimentation, but they are later automated within an ETL (Extract, Transform, Load) pipeline to ensure consistency, reproducibility, and scalability across repeated analyses and production environments.

The primary outputs of the Data Preparation phase include the following artifacts:

1. **Selecting Data (Feature Selection):** Determine which variables and records are retained for modeling.
    - **Inclusion/Exclusion Report:** Document which data elements were retained or removed and justify these decisions using insights from the Data Understanding phase.
2. **Cleaning Data:** Address data quality issues identified earlier.
    - **Data Cleaning Report:** Record actions taken to improve data quality, including:
        - Imputing missing values.
        - Handling outliers, such as capping or removing extreme values.
        - Correcting or transforming skewed distributions.
3. **Creating New Attributes and Records:** Enhance the dataset through feature engineering and record generation.
    - **Derived Attributes List (Feature Engineering):** Describe new attributes created by combining or transforming existing variables.
    - **Generated Records List:** Document any new records created to represent missing, zero-activity, or synthetic cases.
4. **Data Integration:** Combine data from multiple sources into a unified dataset.
    - **Merged Data Report:** Document how datasets were joined or merged, using diagrams when appropriate.
    - **Aggregations Report:** Summarize multiple records into higher-level metrics.
        - Examples of aggregation include:
            - Total purchases per customer.
            - Average purchase value.
            - Percentage of orders paid with credit cards.

Why the Data Preparation Phase Matters

The Data Preparation phase is critical because model performance is tightly coupled to data quality and structure. Effective preparation reduces computational overhead, improves predictive accuracy by emphasizing relevant features, and prevents downstream errors caused by poor data quality.

Carefully documenting preparation decisions ensures transparency, reproducibility, and a clear rationale for the modeling results that follow.

Class Discussion
Can aggressive data preparation improve model performance while simultaneously reducing how well the model represents reality? Where should data scientists draw the line between useful transformation and distortion?

1.5Phase 4: Modeling

Building the Model: The Modeling Phase

In the modeling phase The fourth phase of the CRISP-DM, which involves applying statistical and machine learning algorithms to prepared data in order to predict, classify, or discover patterns in an outcome of interest., mathematical and computational techniques are applied to the prepared dataset to create candidate models. This phase typically involves selecting algorithms, tuning parameters, and iteratively refining data preparation decisions in response to model behavior.

- Selecting and calibrating appropriate modeling algorithms.
- Optimizing parameters and hyperparameters to improve performance.
- Iterating between modeling and data preparation to meet algorithm-specific requirements.

Because different algorithms impose different assumptions and data requirements, close coordination between the Data Preparation and Modeling phases is expected. It is common for modeling results to reveal the need for additional transformations, feature engineering, or data filtering before acceptable performance is achieved.

This phase also introduces the risk of **overfitting**, which occurs when a model learns noise or idiosyncrasies in the training data rather than generalizable patterns. Overfitting is especially likely during repeated experimentation and parameter tuning and must be carefully managed through disciplined testing and validation strategies.

Key Outputs of the Modeling Phase

1. **Modeling Technique Selection**

    - **Selected Candidate Techniques**: Identify the best-performing algorithms under current business, data, and computational constraints.
    - **Record of Assumptions**: Document assumptions associated with each technique, including:
        - Data distribution requirements.
        - Feature independence or multicollinearity considerations.
        - Strategies for handling missing or sparse data.
2. **Test Design Specification**
    - **Design Description**: Specify how data is allocated for training, testing, and validation, including resampling or cross-validation strategies.
3. **Trained Model Artifacts**
    - **Parameter Settings**: Record optimized parameters and provide justification for their selection.
    - **Report of Tested Models**: Document a small set of top-performing candidate models evaluated during experimentation.
        - Model coefficients, weights, or learned structures.
        - Associated parameter configurations.
    - **Model Descriptions**: Summarize findings for each candidate model, including:
        - Relative importance or influence of features.
        - Observed issues such as masking effects from correlated variables.
        - Insights that inform future modeling or data collection efforts.
4. **Preliminary Model Performance Summary**
    - **Performance Metrics**: Report preliminary performance using appropriate measures such as:
        - **Accuracy**: Proportion of correct predictions.
        - **Coefficient of Determination (R^2)**: Proportion of variance explained.
        - **Area Under the Curve (AUC)**: Classification performance across thresholds.

At this stage, performance metrics are used to compare candidate models rather than to make final acceptance decisions. Formal evaluation against business objectives occurs in the Evaluation phase.

Example: Statistical Inference from Modeling

Suppose you are building a model to predict which products customers are likely to purchase at an online bicycle retailer. After testing multiple algorithms, the following patterns may emerge:

- Income and purchase history strongly influence purchasing behavior.
- Gender and ethnicity contribute little explanatory power.
- High correlation between age and income introduces masking effects.

These findings not only inform feature selection and model refinement but also provide actionable business insight, such as identifying which customer attributes should guide marketing or personalization strategies.

Why Model Documentation Matters

Comprehensive documentation during the Modeling phase supports multiple downstream objectives:

- Transparency, allowing stakeholders to understand modeling decisions.
- Reproducibility, enabling future teams to replicate or extend results.
- Governance and compliance, supporting auditing, monitoring, and responsible deployment.

Well-documented models are easier to evaluate, deploy, monitor, and update, reducing long-term risk and increasing organizational trust in analytical systems.

Class Discussion
Should a business ever choose a less accurate model because it is easier to interpret or explain? In what situations might interpretability be more valuable than predictive performance?

1.6Phase 5: Evaluation

Finalizing the Model: The Evaluation Phase

The evaluation phase The fifth phase of the CRISP-DM, which focuses on determining whether candidate models meet technical, business, and operational criteria required to justify deployment. ensures that selected models perform well not only from a technical perspective but also in terms of business value, feasibility, and risk. This phase involves a structured review of modeling results, assumptions, and constraints to determine whether the solution should proceed to deployment, be refined further, or be discontinued.

While the Modeling phase answers the question of which techniques perform best technically, the Evaluation phase answers a different and more consequential question: should the organization move forward with this solution?

Key Objectives of the Evaluation Phase

- **Validate Business Alignment:** Confirm that the model meaningfully addresses the original business problem and supports organizational goals.
- **Assess Practical Feasibility:** Ensure that required data, infrastructure, and expertise remain available, affordable, and sustainable.
- **Quantify Business Impact:** Estimate how the model is expected to influence revenue, costs, risk, or operational performance.

Example: The Netflix Prize

In 2009, Netflix launched a competition offering $1,000,000 to improve its movie recommendation algorithm by 10%. Interim prizes of $50,000 were awarded for smaller 1% improvements. This structure reflected a deliberate evaluation strategy: Netflix explicitly defined performance thresholds that justified further investment.

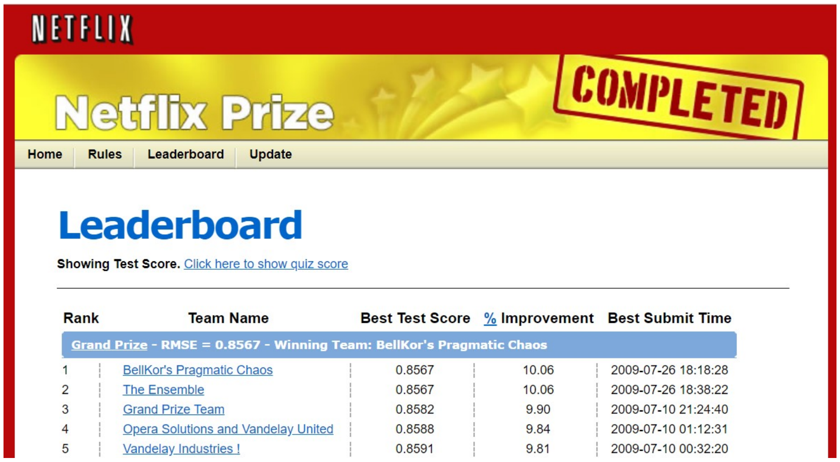


Figure 1.6: Final Leaderboard for the 2009 Netflix Prize

This example highlights an important evaluation principle: performance improvements exhibit diminishing returns. At some point, additional accuracy gains no longer justify their cost, complexity, or implementation risk. Evaluation establishes these decision gates.

A common mistake during this phase is metric fixation, where small improvements in technical metrics are pursued without sufficient consideration of business impact. Effective evaluation balances quantitative performance with strategic judgment.

Outputs of the Evaluation Phase

1. **Evaluation Results Summary:** Synthesize technical performance and expected business impact.
    - **Assessment of Results:** Evaluate how well candidate models meet original business objectives, such as increasing revenue, reducing costs, or improving operational efficiency.
    - **Approved Model Candidates:** Provide a ranked list of models that satisfy minimum criteria to proceed.
2. **Deployment Readiness Review:** Assess organizational and technical readiness.
    - **Process Review:** Evaluate feasibility by examining:
        - Availability and cost of required data inputs.
        - Timeliness and reliability of prediction generation.
        - Potential disruptions to existing business processes.
        - Risks or issues not fully addressed in earlier phases.
3. **Decision and Action Plan:** Define the approved path forward.
    - **List of Actions:** Specify next steps, which may include:
        - Deploying the selected model.
        - Refining the model or collecting additional data.
        - Canceling the project when costs, risks, or limitations outweigh benefits.
    - **Final Decision:** Obtain formal stakeholder approval for the chosen course of action.

Importantly, deciding not to proceed with deployment can represent a successful evaluation outcome. By identifying misalignment or infeasibility early, organizations avoid costly implementation failures.

Why the Evaluation Phase Matters

The Evaluation phase ensures that technical success translates into business value. By integrating performance metrics, feasibility analysis, and strategic judgment, this phase mitigates risk and maximizes the return on data-driven initiatives.

Class Discussion
If a model demonstrates strong technical performance but raises concerns related to cost, risk, or ethics, who should have the authority to stop the project? Should business leaders be able to override data scientists, or vice versa?

1.7Phase 6: Deployment

Deploying the Model: The Deployment Phase

The Deployment Phase The sixth phase of the CRISP-DM process, in which analytical results are delivered as decision-support artifacts or operationalized as models embedded within production systems. ensures that the outputs of a data mining project are translated into usable, accessible, and actionable forms. Deployment may involve sharing insights through reports or dashboards, or integrating predictive models into live systems that automate or support decisions.

Importantly, deployment is not the end of a data project. Instead, it marks the beginning of an operational lifecycle in which models must be monitored, maintained, and periodically reassessed as business conditions and data evolve.

Key Objectives of the Deployment Phase

1. **Deliver Usable Insights:** Present findings and model outputs in formats that support decision-making by stakeholders such as managers, executives, or frontline staff.
2. **Operationalize Models:** Embed predictive models into information systems that automate or augment decisions, such as recommendation engines or risk-scoring tools.

3. **Enable Ongoing Learning:** Capture new data generated by deployed systems and use it to periodically retrain and refine models while maintaining governance and validation controls.

For example, Amazon's product recommendation systems use deployed models to predict customer preferences based on behavioral data. These predictions guide actions while recording new interactions, such as clicks and purchases, which later inform model updates.

Outputs of the Deployment Phase

1. **Deployment Plan:** Define how model outputs will be delivered and consumed.
    - Common deployment approaches include:
        - **Reports:** Static reports or presentations used for strategic or tactical decision-making.
        - **Integrated Systems:** Models embedded within applications or accessed through APIs or web services.
    - For example, students in this course will learn to create web services that allow predictive models to be accessed by applications such as Microsoft Excel.
2. **Monitoring and Maintenance Plan:** Establish procedures to ensure continued model effectiveness.
    - Key monitoring considerations include:
        - Sources and quality of new data used for retraining.
        - Frequency of evaluation to detect model drift The gradual degradation of model performance as data patterns change over time..
        - Processes for validating updated models before redeployment.
    - Visual diagrams are often used to clarify data flows, dependencies, and update cycles.
3. **Final Report and Presentation:** Deliver a comprehensive summary of the entire project.
    - Compile a final report synthesizing outputs from all prior CRISP-DM phases.
    - Present key findings, decisions, and recommendations in a structured presentation format.
4. **Deployment Documentation:** Provide technical and operational details required for ongoing use.
    - Documentation typically includes:
        - API endpoints, authentication requirements, and integration examples.
        - Dependencies such as data source credentials and infrastructure configurations.
    - Platforms such as Microsoft Azure Machine Learning, Amazon SageMaker, and Google Cloud AI can generate portions of this documentation automatically.

A critical risk during deployment is silent failure, in which models continue to produce outputs even though data pipelines have broken or input patterns have changed. Without monitoring, these failures may go undetected and cause greater harm than not deploying a model at all.

Deployment is a shared responsibility among data scientists, engineers, and business stakeholders. Clear ownership and accountability are essential to ensure that models are used appropriately and maintained responsibly.

Ethical and risk considerations also become more prominent at deployment. Models may introduce bias, unintended consequences, or compliance risks when exposed to real-world users, making ongoing review and governance critical.

Why the Deployment Phase Matters

Deployment transforms analytical work into real-world impact. By ensuring accessibility, reliability, and ongoing oversight, this phase maximizes the value of data projects while reducing long-term technical, operational, and ethical risk.

Class Discussion
If a deployed model begins to cause harm due to bias, misuse, or unexpected behavior, who should be held accountable, and what actions should be taken? Should some models never be deployed, even if they perform well technically?

1.8Alternative Frameworks

While CRISP-DM is one of the most widely taught and widely adopted data mining process models, it is not the only framework used in practice. Organizations select or adapt methodologies based on industry context, project scale, regulatory constraints, uncertainty, and the need for agility.

In real-world settings, teams rarely follow a single framework rigidly. Instead, they blend elements from multiple methodologies, using CRISP-DM as a conceptual baseline while adopting specialized practices from more focused frameworks.

Agile Data Science

SEMMA, developed by SAS, places heavy emphasis on data preparation and modeling activities. It is most effective when data sources and objectives are already well defined and predictive modeling is the primary goal. SEMMA is highly structured and closely aligned with SAS tooling, but it provides limited guidance for business framing and long-term deployment.

KDD: Knowledge Discovery in Databases

Knowledge Discovery in Databases (KDD), formalized by Fayyad, Piatetsky-Shapiro, and Smyth, emphasizes extracting novel and useful knowledge from data. It is commonly used in research and academic contexts where interpretation and theory-building are prioritized. KDD provides limited operational guidance, reflecting its focus on discovery rather than deployment.

OSEMN: Obtain, Scrub, Explore, Model, Interpret

OSEMN, associated with Hilary Mason and Chris Wiggins, presents a simplified and accessible data science lifecycle. Its clarity makes it useful for small-scale or exploratory projects, but it lacks guidance for governance, deployment, and long-term maintenance required in production environments.

DMAIC: Define, Measure, Analyze, Improve, Control

DMAIC, developed as part of Six Sigma, is a highly structured framework focused on process improvement and optimization. It is effective for projects with well-defined key performance indicators and repeatable processes. However, its rigor can be time-consuming for projects that require extensive exploratory analysis or rapid iteration.

CRISP-ML(Q): CRISP-DM for Machine Learning Quality

CRISP-ML(Q) extends CRISP-DM to address the realities of deployed machine learning systems. It explicitly incorporates quality assurance, risk management, and lifecycle considerations such as monitoring and maintenance. CRISP-ML(Q) is particularly valuable for projects where models are expected to operate continuously and degrade gracefully over time.

ASUM-DM: Analytics Solutions Unified Method

ASUM-DM, developed by IBM, is an enterprise-oriented extension of CRISP-DM. It provides more detailed guidance, templates, and governance structures for analytics projects implemented at scale. ASUM-DM is commonly used in large organizations that require consistency, documentation, and auditability.

Engineering and Operations Frameworks

In modern organizations, lifecycle frameworks are often complemented by engineering and operational disciplines that focus on reliability, automation, and governance rather than project structure.

Continuous Delivery for Machine Learning (CD4ML) adapts continuous delivery principles to machine learning, emphasizing automated pipelines for training, testing, and deploying models. CD4ML focuses on reducing deployment risk and accelerating iteration through automation.

DataOps applies DevOps principles to data analytics pipelines, improving reliability, collaboration, and speed in data preparation and reporting workflows.

ModelOps extends governance and lifecycle management to a broad range of decision models, including machine learning, statistical models, and rules-based systems. It is especially relevant in regulated or enterprise environments.

The table below summarizes key tradeoffs among these frameworks and highlights situations where each approach is most effective.

Table 1.2
Comparison of Data Mining Process Models

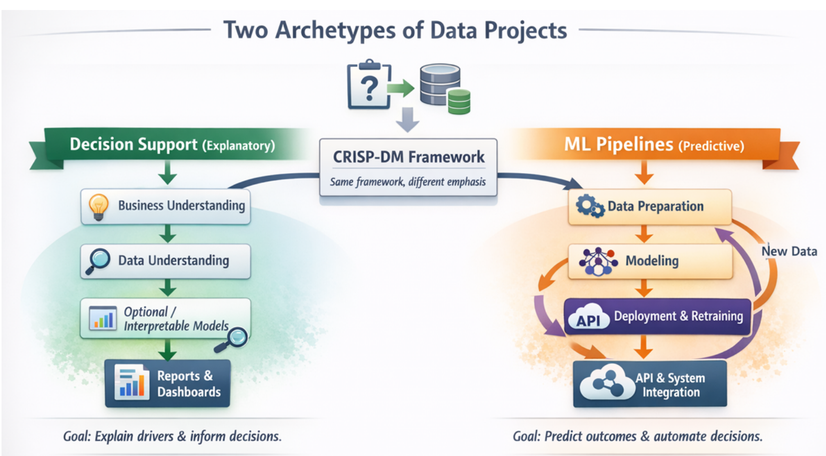
Framework	Business Focus	Flexibility	Collaboration	Deployment Support	Best Use Case
CRISP-DM	High	Moderate	Moderate	Moderate	General-purpose data projects

Framework	Business Focus	Flexibility	Collaboration	Deployment Support	Best Use Case
Agile Data Science	High	High	High	Moderate	Projects with evolving requirements
TDSP	Moderate	Low	High	High	Production-grade machine learning systems
SEMMA	Low	Low	Low	Moderate	Predictive modeling tasks
KDD	Moderate	Moderate	Moderate	Low	Research and knowledge discovery
OSEMN	Low	High	Moderate	Low	Small or exploratory projects
DMAIC	High	Low	Moderate	Moderate	Process improvement initiatives

In this book, we follow CRISP-DM because it provides a clear and generalizable structure for learning how data projects progress from business understanding through deployment. In practice, effective data professionals adapt their approach by blending frameworks and operational practices to fit the project context. Learning one framework deeply makes it easier to understand and apply others.

Class Discussion
How would your choice of methodology differ for a startup experimenting with a new product versus a regulated organization deploying machine learning in healthcare or finance? When does structure improve outcomes, and when does it hinder innovation?

1.9Types of Data Projects



Data mining projects can produce a wide variety of deliverables, ranging from descriptive reports to fully automated machine learning systems. Although these projects differ in complexity, scope, and risk, most can be classified into one of two fundamental categories based on their primary objective: explaining outcomes or predicting outcomes.

Correctly identifying which type of project you are working on is critical because it determines which CRISP-DM phases receive the most emphasis, which methodologies are most appropriate, and how results are ultimately delivered to stakeholders.

The Two Fundamental Types of Data Projects

1. **Decision Support Projects (Explanatory Analytics)**
   - **Primary Goal:** Explain what factors influence an outcome or key performance indicator.
   - **Analytical Focus:** Interpretation, feature influence, and insight generation.
   - **Typical Deliverable:** A report or dashboard that informs human decision-making.
   - **Common Characteristics:**
     - Often uses cross-sectional or snapshot data.
     - Modeling is optional and typically emphasizes interpretability rather than accuracy.
     - Results are consumed by managers, analysts, or executives.
2. **Machine Learning Pipelines (Predictive Analytics)**
   - **Primary Goal:** Predict future outcomes and automate decisions.
   - **Analytical Focus:** Predictive accuracy, scalability, and reliability over time.
   - **Typical Deliverable:** A deployed model accessed through an application or API.
   - **Common Characteristics:**
     - Automates the CRISP-DM lifecycle, including data ingestion, preparation, modeling, evaluation, and deployment.
     - Continuously incorporates new data through scheduled retraining.
     - Operates in real time or near real time with higher operational risk.

Many real-world projects begin as decision support analyses to understand drivers, feasibility, and business value and later evolve into machine learning pipelines once automation is justified. The distinction between these project types reflects differences in usage, risk, and deployment rather than differences in mathematical sophistication.

Adapting CRISP-DM to Different Project Types

CRISP-DM is intentionally flexible and can be applied to both explanatory and predictive projects. However, the emphasis placed on each phase differs substantially depending on the project type. The table below illustrates how the same CRISP-DM framework is adapted for decision support projects versus machine learning pipelines.

Table 1.3
Adapting CRISP-DM for Decision Support and ML Pipelines

Aspect	Decision Support	ML Pipelines
Data Requirements	Uses a static dataset or periodic snapshots; data exploration focuses on understanding features and relationships.	Requires live, continuously updated datasets; automation ensures data preparation adapts to changes over time.
Data Preparation	Includes cleaning, handling outliers, and transformations; focuses on interpretability.	Similar preparation steps, but processes must handle evolving data distributions and unexpected input scenarios.

Aspect	Decision Support	ML Pipelines
Modeling	Optional; often limited to simple models to explain feature impacts.	Essential; dynamically selects the best models and hyperparameters for optimal accuracy over time.
Evaluation Focus	Prioritizes understanding feature importance and standardized coefficients.	Focuses on overall predictive accuracy while minimizing overfitting and removing features with low predictive value.
Deployment	Delivers static reports or dashboards; informs management-level decisions.	Deploys a predictive model as an API; schedules retraining to ensure up-to-date, real-time predictions.
Performance Monitoring	Rarely performed beyond periodic updates to repeat the analysis.	Continuously monitors prediction accuracy and model drift; revises models as data patterns evolve.

Frameworks and Methodologies by Project Type

Different data project methodologies align more naturally with different project types. Decision support projects emphasize interpretation and communication, while machine learning pipelines emphasize engineering, automation, and governance.

- **Decision Support Projects:** CRISP-DM, OSEMN, KDD, SEMMA
- **Machine Learning Pipelines:** TDSP, CRISP-ML(Q), Agile Data Science combined with MLOps, CD4ML, and ModelOps

If you ever find yourself asking, “Why are we doing this step?” the answer can almost always be traced back to which type of project you are working on and which CRISP-DM phase requires the most attention.

1.10Reading Quiz

Take the quiz below to make sure you remember each of the key terms and concepts in this chapter.

This assessment can be taken online.

<{http://www.bookeducator.com/Textbook}learningobjective >Explain the phases of the Cross-Industry Standard for Data Mining